THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

032/2C CHEMISTRY 2C

(Actual Practical)

TIME: 2:30 HOURS **YEAR: 2025**

INSTRUCTIONS

1. This paper consists of **two (2)** questions.
2. Answer **all** questions.
3. Each question carries **twenty-five (25)** marks
4. Non programming calculators may be used.
5. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer sheets provided.
8. You may use the following constants:-

   Atomic masses: H = 1, O = 16, Na= 23, C=12, S =32, Cl = 35.5

   1litre = 1dm3 = 1000cm3

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1. You are provided with the following solutions;

   - Solution **X**: Containing 0.1M of hydrochloric acid.
   - Solution **Y**: Containing 2.65g of G_2CO_3 per 0.5L of solution.
   - Methyl orange indicator

Procedure:

(a) Put the solution **X** into the burette. Pipette 20 ml or 25ml of solution **Y** into a titration flask.

(b) Add three drops of methyl orange indicator.

(c) Titrate this solution **X** against the solution **Y** until the end point is reached.

(d) Repeat the titration to obtain three more values and record your results as shown in tabular form.

(e) (i) Draw the table of results and complete it by filling the titre values.

(ii) Calculate the average volume of acid used.

(f) The colour change at the end point was from_to____.

(g) Calculate the molarity of Solution **Y**.

(h) Determine the atomic mass of **G** in G_2CO_3 and name it.

(i)

2. You are provided with the following
   - Solution A containing 0.5M sodium thiosulphate
   - Solution B containing 0.1M sulphuric acid
   - A piece of paper marked 'X' and stop watch.

Procedure

(a) Using a measuring cylinder measure 10ml of solution A and pour it into the 50ml beaker.

(b) Measure 10ml of solution B using different measuring cylinder.

(c) Pour solution B into the beaker containing solution A and start the stop watch immediately.

(d) Swirl the flask twice and observe the cross 'x' on the marked paper until it disappears.

(e) Record the time. Through away the contents of the beaker and rinse it.

(f) Repeat the procedure using the volume of A and mix with distilled water as shown below;

Table 1

Volume of A, V_A(ml)	Volume of distilled water, V_w (ml)	Volume of B, V_B (ml)	Time (s)	$Rate$, $\frac{1}{t}$(s^{-1})
10	0	10		
8	2	10		
6	4	10		
4	6	10		
2	8	10		

(g) Plot the graph of:

(i) Volume of $Na_2S_2O_3$(ml) versus time(s).

(ii) Volume of $Na_2S_2O_3$(ml) versus rate (s^{-1}).

(h) From the experiment you attempted, what conclusion can you draw?